Boxplots

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Introduction

A boxplot — also called a box-and-whisker plot — compresses a continuous distribution into a five-number summary: minimum, first quartile, median, third quartile, maximum, with Tukey's fences at $\pm 1.5 \times \text{IQR}$ defining the whiskers and flagging anything beyond as an outlier. Boxplots are the workhorse for compact group-by-group comparisons because they put many groups side by side without ink overload. They are also the geom most often misused: they hide bimodality, hide sample size, and hide the raw observations behind a tidy rectangle, all of which can mislead readers who do not realise what the box actually summarises.

Prerequisites

A working understanding of quartiles, the interquartile range, and the standard $\pm 1.5 \cdot \text{IQR}$ outlier convention.

Theory

The standard Tukey boxplot has the following components per group:

- A box spanning the first and third quartiles (the IQR).
- A horizontal line at the median inside the box.
- Whiskers extending to the most extreme observation within $\pm 1.5 \times \text{IQR}$ of the box edges.
- Individual points beyond the whiskers drawn as outliers.

Variants:

- **Notched boxplot**: notches at $\pm 1.57 \cdot \text{IQR}/\sqrt{n}$ give an approximate 95 % confidence interval for the median; non-overlapping notches between groups suggest a real difference in medians.
- **Variable-width boxplot**: box width proportional to $\sqrt{n}$, encoding group size visually.
- **Letter-value plot**: replaces the single box with nested boxes of decreasing depth, suitable for very large n where a single Tukey box hides tail structure.

Assumptions

For descriptive use, none. For the notched-boxplot CI interpretation, the approximation assumes approximately symmetric within-group distributions; under heavy skew the notches mislead.

R Implementation

```r
library(ggplot2)

# Standard
ggplot(mtcars, aes(factor(cyl), mpg, fill = factor(cyl))) +
  geom_boxplot() +
  scale_fill_brewer(palette = "Set2") +
  theme_minimal()

# Notched
ggplot(iris, aes(Species, Sepal.Length, fill = Species)) +
```

```r
  geom_boxplot(notch = TRUE) +
  scale_fill_brewer(palette = "Set2") +
  theme_minimal()

# Boxplot + jittered points (show data and summary)
ggplot(mtcars, aes(factor(cyl), mpg)) +
  geom_boxplot(outlier.shape = NA, fill = "#2A9D8F", alpha = 0.5) +
  geom_jitter(width = 0.2, alpha = 0.5) +
  theme_minimal()
```

Output & Results

Three views: a standard side-by-side boxplot of mpg by cylinder count, a notched boxplot of iris sepal length per species (notches show approximate median CIs), and a hybrid plot that hides the default outlier dots and overlays jittered raw points so the boxplot's summary and the underlying data are both visible.

Interpretation

A reporting sentence (figure caption): "Boxplots of fuel efficiency by cylinder count (n = 11, 7, 14 for 4, 6, 8 cylinders); boxes span Q1 to Q3, the line marks the median, whiskers extend to the most extreme observation within $1.5 \cdot \text{IQR}$, and dots are outliers." Always describe the whisker definition; some software uses 5 %/95 % quantiles or min/max instead of the Tukey rule.

Practical Tips

- Hide default outliers (`outlier.shape = NA`) whenever you overlay `geom_jitter()`; otherwise extreme points appear twice.
- Notches can overshoot the box for very small samples; R prints a warning when this happens, and the plot becomes uninterpretable.
- Use `varwidth = TRUE` to encode group sizes via box width when groups are unbalanced; the visual cue is subtle but informative.
- For very small samples ($n < 10$), prefer a dot plot or strip plot over a boxplot; quartile-based summaries are unstable for tiny groups.
- Boxplots hide multi-modality; always inspect a density or histogram alongside before concluding that a group is unimodal.
- For very large samples, a letter-value plot (`lvplot::geom_lv()`) preserves more of the tail structure than a single Tukey box.

R Packages Used

`ggplot2` for `geom_boxplot()` and the `varwidth` / `notch` parameters; `lvplot` for letter-value plots at very large n; `ggbeeswarm` and `ggdist` for richer alternatives that combine boxplot summaries with raw points and distributional shape.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- ggplot2 grammar and multi-panel layouts